

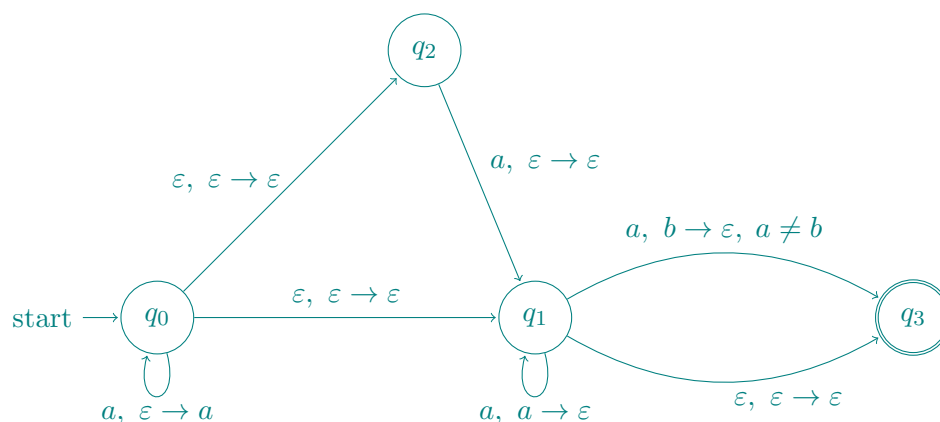
Homework 4: Push Down Automata

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You should submit a typeset or *neatly* written pdf on Gradescope. The grading TA should not have to struggle to read what you've written; if your handwriting is hard to decipher, you will be asked to typeset your future assignments. Three bonus points if you use L^AT_EX, and our template. You may collaborate with other students, but any written work should be your own.

1. Give the state diagram of a PDA for $\{w \in \Sigma^* \mid w \text{ is not a palindrome}\}$

**Answer:**

2. Prove that the context-free languages are not closed under intersection. You may assume that $\{a^n b^n c^n \mid n \in \mathbb{N}\}$ is not a context-free language.

Answer: We will assume CFLs are closed under intersection. With this logic, we observe the intersection of any two CFLs will also be a CFLs. We observe the language $L_1 = \{a^n b^n c^m \mid n, m \geq 0\}$ is a CFL since it can be represented with the following productions: $S \rightarrow aSb \mid C$ and $C \rightarrow cC \mid \epsilon$. Additionally, we can also design a language L_2 such that it is a CFL with the properties $L_2 = \{a^m b^n c^n \mid m, n \geq 0\}$ and productions of $S \rightarrow aS \mid T$ and $T \rightarrow bTc \mid \epsilon$.

We now wish to take the union of both languages. We observe $L_1 \cap L_2 = \{a^n b^n c^n \mid n \geq 0\}$ is not context-free, as discussed in lecture.

Therefore, we find this contradicts our original assumption of closure under intersection. Therefore, we find CFLs are not closed under intersection.

3. Let R be a regular language, and let C be a context-free language. Prove that $R \cap C$ is context-free.

Answer: We define an F such that R is recognized by the finite automaton $F = (S, \Sigma, \delta, s_0, F)$. Similarly, we define a P such that C is recognized by the PDA $P = (Q, \Sigma, \tau, \delta, q_0, Z, F)$. We are able to define a new PDA P' such that P and F are simultaneously simulated by using Cartesian states in the general form (q, s) where $q \in Q$ and $s \in S$.

For each input symbol $a \in \Sigma$ and for every transition $(q, a, \gamma) \rightarrow (p, \alpha)$ in δ , we define a transition in P' in the form $((q, s), a, \gamma) \rightarrow (p, \delta(s, a), \alpha)$. For empty transitions (ε) , we have $((q, s), \varepsilon, \gamma) \rightarrow ((p, s), \alpha)$.

We recognize the starting state of P' is (q_0, s_0) and the accepting state(s) is/are the Cartesian state cooresponding to the accepting states of P and F . Therefore, we find P' accepts an input if, and only if, it is accepted by both P and F , which means the string is in C and R . Finally, since P' is a PDA, it recognizes a CFL. Therefore, the intersection of F and C is context-free.